

SIMATS ENGINEERING

ASSESSMENT TOOL 1

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COURSE NAME: Database Management Systems

REGNO: 192524284

COURSE CODE: CSA05

Q1

Given a set of relations and sample data, write SQL queries to retrieve specific information using joins and subqueries, and explain the logic used. Bloom's Level:

BL3 – Apply

1.CREATE STUDENT TABLE

```
mysql> CREATE TABLE student (
->   RegNo INT PRIMARY KEY,
->   Name VARCHAR(20),
->   Gender CHAR(1),
->   DOB DATE,
->   MobileNo BIGINT,
->   City VARCHAR(20),
->   FacNo INT
-> );
Query OK, 0 rows affected (0.13 sec)

mysql> DESC student;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| RegNo | int  | NO   | PRI | NULL    |       |
| Name  | varchar(20) | YES |     | NULL    |       |
| Gender | char(1) | YES |     | NULL    |       |
| DOB   | date  | YES |     | NULL    |       |
| MobileNo | bigint | YES |     | NULL    |       |
| City  | varchar(20) | YES |     | NULL    |       |
| FacNo | int   | YES |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.10 sec)
```

2.CREATE DEPARTMENT TABLE

```
mysql> CREATE TABLE Department (
->   DeptNo INT PRIMARY KEY,
->   DeptName VARCHAR(20),
->   HOD VARCHAR(20)
-> );
ERROR 1050 (42S01): Table 'department' already exists
mysql> DESC Department;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| DeptNo | varchar(4) | NO | PRI | NULL |       |
| DeptName | varchar(15) | YES |     | NULL |       |
| DeptHead | varchar(4) | YES |     | NULL |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.06 sec)
```

3. Insert Sample Data

Department

```
mysql> INSERT INTO Department VALUES
-> (101, 'CSE', 'Mohan'),
-> (102, 'ECE', 'Raju'),
-> (103, 'EEE', 'Ramesh');
Query OK, 3 rows affected (0.07 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Department;
+-----+-----+-----+
| DeptNo | DeptName | HOD    |
+-----+-----+-----+
| 101    | CSE      | Mohan  |
| 102    | ECE      | Raju   |
| 103    | EEE      | Ramesh |
+-----+-----+-----+
3 rows in set (0.04 sec)
```

Student

```
mysql> INSERT INTO Student VALUES
-> (19221151, 'Akash', 'M', '2004-01-12', 9876543210, 'Chennai', 101),
-> (19221152, 'Priya', 'F', '2005-02-13', 9876543211, 'Madurai', 102),
-> (19221153, 'Rahul', 'M', '2004-03-14', 9876543212, 'Salem', 101),
-> (19221154, 'Sneha', 'F', '2005-04-15', 9876543213, 'Coimbatore', 103),
-> (19221155, 'Kiran', 'M', '2004-05-16', 9876543214, 'Trichy', 102);
Query OK, 5 rows affected (0.05 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> DESC STUDENT;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| RegNo      | int           | NO   | PRI | NULL    |       |
| Name       | varchar(20)   | YES  |     | NULL    |       |
| Gender     | char(1)       | YES  |     | NULL    |       |
| DOB        | date          | YES  |     | NULL    |       |
| MobileNo   | bigint        | YES  |     | NULL    |       |
| City       | varchar(20)   | YES  |     | NULL    |       |
| FacNo      | int           | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.01 sec)
```

JOIN Queries

1. Display Student Name and Department Name

```
mysql> SELECT Student.Name, Department.DeptName  
-> FROM Student  
-> INNER JOIN Department  
-> ON Student.DeptNo = Department.DeptNo;
```

Name	DeptName
Akash	CSE
Priya	ECE
Rahul	CSE
Sneha	EEE
Kiran	ECE

```
5 rows in set (0.00 sec)
```

Q2

Using relational algebra, formulate expressions to solve complex queries such as retrieving records that satisfy multiple conditions across relations. Bloom's Level: BL4 – Analyse

ANS:

Relations

STUDENT(RegNo, Name, Gender, DeptNo)

DEPARTMENT(DepartmentNo, DepartmentName, HOD)

Query 1

Retrieve the names of students who belong to the CSE department.

Relational Algebra:

$\pi_{Name}(\sigma_{DeptName='CSE'}(STUDENT \bowtie STUDENT.DeptNo = DEPARTMENT.DeptNo DEPARTMENT))$

Query 2

Retrieve the names of female students in the CSE department.

Relational Algebra:

$\pi_{Name}(\sigma_{Gender='F' \wedge DeptName='CSE'}(STUDENT \bowtie STUDENT.DeptNo = DEPARTMENT.DeptNo DEPARTMENT))$

Query 3

Retrieve the names of students whose HOD is 'Mohan'.

Relational Algebra:

$\pi_{Name}(\sigma_{HOD='Mohan'}(STUDENT \bowtie STUDENT.DeptNo = DEPARTMENT.DeptNo DEPARTMENT))$

Query 4

Retrieve the names and department names of all students.

Relational Algebra:

$\pi_{\text{Name,DeptName}}$

$(\text{STUDENT} \bowtie \text{STUDENT.DeptNo} = \text{DEPARTMENT.DeptNo DEPARTMENT})$

Query 5

Retrieve the department names having female students.

Relational Algebra:

π_{DeptName}

$(\sigma_{\text{Gender}='F'}$

$(\text{STUDENT} \bowtie \text{STUDENT.DeptNo} = \text{DEPARTMENT.DeptNo DEPARTMENT}))$

Query 6

Retrieve the names of students who belong to the ECE department.

Relational Algebra:

π_{Name}

$(\sigma_{\text{DeptName}='ECE'}$

$(\text{STUDENT} \bowtie \text{STUDENT.DeptNo} = \text{DEPARTMENT.DeptNo DEPARTMENT}))$

Operators Used

Symbol Meaning

- | | |
|-----------|--|
| σ | Selection (filters rows based on a condition) |
| π | Projection (selects required columns) |
| $\bowtie$ | Join (combines two relations using a common attribute) |
| $\wedge$ | Logical AND (multiple conditions) |

Explanation

- Selection (σ) is used to filter records based on one or more conditions.
- Projection (π) is used to retrieve only the required attributes.

- Join (⋈) combines the STUDENT and DEPARTMENT relations using the common attribute DeptNo.

Bottom of Form

Q3

relational schema and construct appropriate SQL queries for data retrieval and updates. Bloom's Level: BL6 – Create

1. Create Student Table

```
mysql> CREATE TABLE student (
->   RegNo INT PRIMARY KEY,
->   Name VARCHAR(20),
->   Gender CHAR(1),
->   DOB DATE,
->   MobileNo BIGINT,
->   City VARCHAR(20),
->   FacNo INT
-> );
Query OK, 0 rows affected (0.13 sec)

mysql> DESC student;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| RegNo | int           | NO   | PRI | NULL    |       |
| Name  | varchar(20)   | YES  |     | NULL    |       |
| Gender | char(1)       | YES  |     | NULL    |       |
| DOB   | date         | YES  |     | NULL    |       |
| MobileNo | bigint      | YES  |     | NULL    |       |
| City  | varchar(20)   | YES  |     | NULL    |       |
| FacNo | int           | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.10 sec)
```

2. Insert Sample Data

```
mysql> CREATE TABLE faculty
-> (
->   RegNo INT(3),
->   Name VARCHAR(15),
->   Gender CHAR(1),
->   DOB DATE,
->   MobileNo BIGINT,
->   City VARCHAR(15)
-> );
ERROR 1050 (42S01): Table 'faculty' already exists
mysql> INSERT INTO faculty
-> VALUES
-> (11911511,'mohan','m','2004-01-12',998826973,'mtr'),
-> (11922112,'raju','m','2005-02-13',908658694,'mgr'),
-> (11922113,'ramesh','m','2005-03-14',998263548,'mnr'),
-> (11922154,'ramu','m','2006-04-14',908269279,'mgr'),
-> (11922115,'san','m','2007-05-17',998269245,'hyt');
Query OK, 5 rows affected (0.06 sec)
Records: 5 Duplicates: 0 Warnings: 0

mysql> desc faculty;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| RegNo | int           | YES  |     | NULL    |       |
| Name  | varchar(15)   | YES  |     | NULL    |       |
| Gender | char(1)       | YES  |     | NULL    |       |
| DOB   | date         | YES  |     | NULL    |       |
| MobileNo | bigint      | YES  |     | NULL    |       |
| City  | varchar(15)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)
```

3. Display All Students

```
mysql> SELECT * FROM Student;
```

RegNo	Name	Gender	DOB	MobileNo	City
19221151	Mohan	M	2004-01-12	990826973	Madurai
19221152	Raju	M	2005-02-13	908658694	Chennai
19221153	Ramesh	M	2005-03-14	998263548	Trichy
19221154	Ram	M	2006-04-14	908269279	Salem
19221155	San	M	2007-05-17	998269245	Coimbatore

```
5 rows in set (0.28 sec)
```

4. Display students from Chennai.

```
mysql> SELECT *  
-> FROM Student  
-> WHERE City='Chennai';
```

RegNo	Name	Gender	DOB	MobileNo	City
19221152	Raju	M	2005-02-13	908658694	Chennai

```
1 row in set (0.06 sec)
```

Q4. Analyse a given SQL query with nested subqueries and predict its output, explaining each step of execution. Bloom's Level: BL4 – Analyze

Nested Subquery

```
mysql> SELECT * FROM Department;
```

DeptNo	DeptName	HOD
101	CSE	Mohan
102	ECE	Raju
103	EEE	Ramesh

```
3 rows in set (0.00 sec)
```

Q5 Compare different SQL approaches (joins vs subqueries) for solving the same problem and evaluate their efficiency. Bloom's Level: BL5 – Evaluate

ANS;

```
mysql> SELECT s.Name, d.DeptName
-> FROM Student s
-> INNER JOIN Department d
-> ON s.DeptNo = d.DeptNo;
```

Name	DeptName
Mohan	CSE
Raju	ECE
Ramesh	CSE
Ram	EEE
San	ECE

```
5 rows in set (0.05 sec)
```


Q6 .Design and implement views for a database system to provide restricted data access, and justify their use for security and abstraction.Bloom's Level: BL6 – Create

ANS;

```
mysql> CREATE VIEW Student_View AS  
-> SELECT RegNo, Name, City  
-> FROM Student;  
Query OK, 0 rows affected (0.12 sec)
```

```
mysql> SELECT * FROM Student_View;  
+-----+-----+-----+  
| RegNo   | Name   | City     |  
+-----+-----+-----+  
| 19221151 | Mohan  | Madurai  |  
| 19221152 | Raju   | Chennai  |  
| 19221153 | Ramesh | Trichy   |  
| 19221154 | Ram    | Salem   |  
| 19221155 | San    | Coimbatore |  
+-----+-----+-----+  
5 rows in set (0.03 sec)
```

Q7 Given a database schema, write relational algebra expressions and convert them into equivalent SQL queries, analyzing differences in representation. Bloom's Level: BL4 – Analyze

ANS; Given Database Schema

Student

Student_ID	Student_Name	Dept_ID	Marks
------------	--------------	---------	-------

Department

Dept_ID	Dept_Name
---------	-----------

Example 1: Students in the CSE Department

Relational Algebra

$\pi_{\text{Student_Name}, \text{Dept_Name}}(\sigma_{\text{Dept_Name} = \text{'CSE'}}(\text{Student} \bowtie \text{Department}))$

Equivalent SQL

SQL

SELECT Student.Student_Name, Department.Dept_Name

FROM Student

JOIN Department

ON Student.Dept_ID = Department.Dept_ID

WHERE Department.Dept_Name = 'CSE';

Analysis

- **Relational Algebra: Uses σ (Selection), $\bowtie$ (Join), and π (Projection).**
- **SQL: Uses WHERE, JOIN, and SELECT.**
- **Both return the same result, but SQL is more user-friendly and executable.**

Example 2: Students Scoring More Than 80

Relational Algebra

$\pi_{\text{Student_Name, Marks}}(\sigma_{\text{Marks} > 80}(\text{Student}))$

Equivalent SQL

SELECT Student_Name, Marks

FROM Student

WHERE Marks > 80;

Analysis

- **RA clearly separates selection and projection.**
- **SQL combines both operations into one query using SELECT and WHERE.**

Example 3: Average Marks Department-wise

Relational Algebra (Extended)

$\gamma_{\text{Dept_ID}, \text{AVG}(\text{Marks}) \rightarrow \text{Avg_Marks}}(\text{Student})$

Equivalent SQL :

```
SELECT Dept_ID,  
AVG(Marks) AS Avg_Marks  
FROM Student  
GROUP BY Dept_ID;
```

Analysis

- Extended RA uses the γ (Grouping) operator.
- SQL uses GROUP BY with aggregate function
- SQL is easier to read and directly supported by DBMSs.

Example 4: Student Names with Department Names

Relational Algebra

$\pi_{\text{Student_Name}, \text{Dept_Name}}(\text{Student} \bowtie \text{Department})$

Equivalent SQL:

```
SELECT Student.Student_Name,  
Department.Dept_Name
```

FROM Student

JOIN Department

ON Student.Dept_ID = Department.Dept_ID;

Analysis

- **RA uses the join operator $\bowtie$.**
- **SQL uses the JOIN ... ON clause.**
- **Both produce the same output.**

Comparison: Relational Algebra vs SQL

Relational Algebra	SQL
---------------------------	------------

Procedural query language	Declarative query language
----------------------------------	-----------------------------------

Uses symbols like σ, π, $\bowtie$	Uses keywords like SELECT, FROM, WHERE
--	---

Describes the sequence of operations	Describes the required result
---	--------------------------------------

Mainly used for query design and optimization	Used to interact with real databases
--	---

Theoretical foundation	Practical implementation
-------------------------------	---------------------------------

Database Schema

STUDENT(RegNo, Name, Gender, DOB, MobileNo, City)

Query 1

Display the names of students from Chennai.

Relational Algebra

$\pi_{\text{Name}}(\sigma_{\text{City}=\text{'Chennai'}}(\text{STUDENT}))$

```
mysql> SELECT Name
-> FROM Student
-> WHERE City='Chennai';
```

Name
Raju

```
1 row in set (0.01 sec)
```

Query 2

Display the registration number and name of male students.

Relational Algebra

$\pi_{\text{RegNo}, \text{Name}}(\sigma_{\text{Gender}='M'}(\text{STUDENT}))$

```
mysql> SELECT RegNo, Name
-> FROM Student
-> WHERE Gender='M';
```

RegNo	Name
19221151	Mohan
19221152	Raju
19221153	Ramesh
19221154	Ram
19221155	San

```
5 rows in set (0.00 sec)
```

Query 3

Display all students born after 01-01-2005.

Relational Algebra

$\sigma_{\text{DOB} > '2005-01-01'}(\text{STUDENT})$

```
mysql> SELECT *
-> FROM Student
-> WHERE DOB > '2005-01-01';
```

RegNo	Name	Gender	DOB	MobileNo	City	DeptNo
19221152	Raju	M	2005-02-13	988658694	Chennai	102
19221153	Ramesh	M	2005-03-14	988263548	Trichy	101
19221154	Ram	M	2006-04-14	988269279	Salem	103
19221155	San	M	2007-05-17	998269245	Coimbatore	102

```
4 rows in set (0.06 sec)
```

Q8 .Optimize a set of inefficient SQL queries by restructuring them using joins, indexing concepts, or views, and evaluate performance improvements.

Bloom's Level: BL5 – Evaluate

ANS: Query 1: Inefficient Query (Using Subquery)

Problem: Display the names of students who belong to the CSE department.

```
mysql> SELECT Name
-> FROM Student
-> WHERE DeptNo =
-> (
-> SELECT DeptNo
-> FROM Department
-> WHERE DeptName = 'CSE'
-> );
```

Name
Mohan
Ramesh

2 rows in set (0.06 sec)

Query 2: Optimization Using View

Create a view to simplify repeated queries.

```
mysql> CREATE VIEW Student_Department AS
-> SELECT s.RegNo,
->        s.Name,
->        d.DeptName
-> FROM Student s
-> INNER JOIN Department d
-> ON s.DeptNo = d.DeptNo;
Query OK, 0 rows affected (0.03 sec)

mysql> SELECT * FROM Student_Department;
```

RegNo	Name	DeptName
19221151	Mohan	CSE
19221153	Ramesh	CSE
19221152	Raju	ECE
19221155	San	ECE
19221154	Ram	EEE

5 rows in set (0.06 sec)

Q9. Develop a complex SQL query involving grouping, aggregation, and nested queries to solve a real-world problem

Scenario

A Student Management System stores student records. The administrator wants to know the number of students in each city and identify cities that have more than one student.

Query 1: Grouping and Aggregation

Display the number of students in each city.

```
mysql> SELECT City, COUNT(*) AS Total_Students  
-> FROM Student  
-> GROUP BY City;
```

City	Total_Students
Madurai	1
Chennai	1
Trichy	1
Salem	1
Coimbatore	1

5 rows in set (0.06 sec)

Q10. Given multiple relations, analyze the query requirements and decide whether to use Views, JOINS, or Subqueries, justifying your choice with reasoning.

Bloom's Level: BL5 – Evaluate

Relations

STUDENT(RegNo, Name, Gender, DeptNo, City)

DEPARTMENT(DeptNo, DeptName, HOD)

1. Using JOIN

Query

Display the student name along with the department name.


```
mysql> SELECT s.RegNo, s.Name, d.DeptName
-> FROM Student s
-> INNER JOIN Department d
-> ON s.DeptNo = d.DeptNo;
```

RegNo	Name	DeptName
19221151	Mohan	CSE
19221153	Ramesh	CSE
19221152	Raju	ECE
19221155	San	ECE
19221154	Ram	EEE

5 rows in set (0.00 sec)

Reason

- JOIN is used because data is required from both Student and Department tables.
- It is faster and more efficient for retrieving related data from multiple tables.

2. Using SUBQUERY

- Query
- Display the names of students who belong to the CSE department.

```
mysql> SELECT Name
-> FROM Student
-> WHERE DeptNo =
-> (
-> SELECT DeptNo
-> FROM Department
-> WHERE DeptName = 'CSE'
-> );
```

Name
Mohan
Ramesh

2 rows in set (0.05 sec)

